Many Worlds interpretation of quantum mechanics, 169n
matter
  bending of space by, 68, 108
  Big Bang Nucleosynthesis and, 45
  dark energy and, 107–8
  density over time of, 97 (fig.)
  imbalance between antimatter and, 162
  quark era and distinction between antimatter and, 44
maximum entropy universe, 97
McNees, Robert, 153n
Mendeleev, Dmitri, 137
Mercury
  gravity theory and observations of, 8, 162
  red giant phase of the Sun and destruction of, 1, 121
Milky Way, 8, 24
  Andromeda Galaxy's future collision with, 51–52, 62–63, 88
  distance ladder measurements and, 119 (fig.)
  smaller nearby galaxies consumed by, 52
"Mixed Signals" (White and Wharton), 166
Moon
  Big Rip and, 113
  cosmic ray collisions on, 132–33
  distance and apparent size of, 86
Moss, Ian, 150

neutron, decay of, 89
neutron stars, 125n, 158, 159, 166, 171
Newton, Sir Isaac, 8, 9
Nietzsche, Friedrich, 3, 100–101

observable universe
  description of, 21, 26
  cartoon map of, 26 (fig.)
  cosmic expansion and, 54
  particle horizon in, 82–84
  seeing "edge" of, 83
  uniformity problem of, 39–40